UM-2: a 4+4 model of the Universe

Einstein-Lovelock gravity, the dirac16complex field, photons, neutrinos, electrons, up and down quarks, gluons, and the W^+, W^- and Z bosons without a Higgs particle; every mass from the x_0 dependence of wave functions

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1 How to read this document

This document DEFINES the model UM-2. Every field, every interaction and every equation of the model is written out here; nothing has to be looked up elsewhere. Each statement carries one of these labels:

- **DERIVED**: computed and verified in this work (sympy or numerics; the verification is stated with the result);
- **AUTHOR**: taken from the author's notebooks (coordinates, metric, Lovelock weights, gamma matrices, mass parameter);
- **STANDARD**: textbook physics used as an input;
- **ASSUMED**: a choice made for this model;
- **INVENTED**: a new phenomenological ingredient of this model;
- **OPEN**: not resolved in this version.

Units: $\hbar = c = 1$ unless stated. Indices $M, N, \ldots$ run over the eight coordinates, $a, b, \ldots$ over the orthonormal frame, μ, ν over the four observable coordinates x_1, x_2, x_3, x_4.

Symbols with more than one use:

- k is the block variable $1/f^2$ (k_*, $k_\pm$, δk, $G(k)$); as the index of α_k, $E_{(k)}$ and $\mathcal{L}_{(k)}$ it is the Lovelock order (written p in the reduction formula of section 8.5, where the block variable appears as well); in e^{ikx_1}, in h_j, in $\vec{k}$, in Δk and in the dispersion $\varepsilon = e^{-\alpha_4}|k|$ it is the momentum of a mode along x_1, x_2, x_3.
- $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and Λ_4 the four-dimensional one; in the x_0 eigenvalue problems $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue (Λ_n, Λ_W, Λ_Z, Λ_D are particular ones); $\Lambda_\star$ is the scale of the effective interactions; Λ_ℓ and Λ_q are the tip coupling matrices.
- n is a number density (n_o that of a Kohn-Sham orbital), except as a level or period number (Λ_n, $z \to n\pi$) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell.
- The subscript t (x_t, p_t, v_t) refers to the extra times x_5, x_6, x_7; $\xi = y/L$ is the scaled hidden coordinate.
- f is the block scale and f_o an occupation number. ρ is an energy density, except for the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$; ϱ is a probability density in ξ.
- $w = (w_1, w_2, w_3)$ are the Lovelock weights; w is also the weight $1/2$ of a fold parity in the Kohn-Sham sums, $\sqrt{c^2 - e^2}$ in the fermion mass condition, and the equation-of-state parameter p_3/ρ. $z = 6Hx_0$, except where it is a redshift (dark energy, cosmology).

2 Summary

1. **Spacetime.** Eight dimensions with the author's coordinates $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$: x_0 is the hidden space-like direction, x_1, x_2, x_3 are ordinary space, x_4 is the time, and x_5, x_6, x_7 are three extra time-like directions. The frame metric is $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in this order. The metric depends on x_0 only through $z = 6Hx_0$ with period π in z, i.e. period $\ell_0 = \pi/(6H)$ in x_0.
2. **Gravity.** Einstein-Lovelock gravity in eight dimensions with the author's weights $w = (1, 1/32, 1/1152)$ and $\Lambda = -10H^2$, kept unchanged. A block stabiliser (a vacuum energy of the hidden block, ASSUMED) gives a

stable static block at $k_* = 12 + \frac{4}{3}\sqrt{51}$ on which four-dimensional gravity is attractive: its Einstein-Hilbert coefficient is $G(k_*) = \frac{20}{3} + \sqrt{51} = 13.81 > 0$ (DERIVED).

3. **The deflating extra times.** In the primordial era the extra times x_5, x_6, x_7 (and x_0 with them) deflate at rates up to $2.6H$ into the stabilised block, then the block oscillates with the radion frequency $3.387H$ while 3-space keeps expanding (DERIVED, full nonlinear equations).
4. **dirac16complex** Ψ: a 16-component complex anticommuting spinor of $Cl(4,4)$ (Pin(4,4) irreducible; two inequivalent 8-dimensional irreducible representations of Spin(4,4)) coupled to gravity through the canonical spin connection, which is non-trivial: $\gamma^M \Omega_M = \frac{3}{2}(H/f)\gamma^{x_0} + (\dot{b}_4/b_4 - \frac{1}{2}\dot{a}_4)\gamma^{x_4}$. It splits into two Lorentz-invariant halves, Ψ_A ($B = -1$) and Ψ_B ($B = +1$), with $B = -iC\gamma^{x_4}$ the Krein metric.
5. **Standard-Model content (one generation, no Higgs particle).** Photon, gluons, W^+ (charge $+e$), W^- (charge $-e$), Z (neutral), the electron, one neutrino, the up and down quarks (confined into protons and neutrons). Electroweak symmetry is broken at the regular tip of the hidden direction by a non-linear sigma field $\Sigma \in SU(2)$ that has no radial mode, hence no Higgs particle.
6. **Masses.** Every four-dimensional mass is the eigenvalue of an x_0 operator acting on a wave function periodic in x_0 with period $\pi/(6H)$. Massless: graviton, photon, gluons (flat x_0 profiles) and the fold band of Ψ (chiral zero modes). Massive: $W^{\pm}$, Z (profiles bent at the tip), e, ν, u, d (chiral zero modes mixed at the tip), the bulk band of Ψ and every Kaluza-Klein mode (DERIVED spectra, fitted to the measured masses with order-one parameters).
7. **New interactions (INVENTED).** A chiral tip Z portal makes the Ψ_A fold band behave as two extra neutrino species, so that the Z invisible width ($N_\nu = 2.984$) is reproduced and $N_{eff} = 3.044$ follows; a gauge-kinetic portal couples Ψ to pairs of gauge bosons (W^+W^-, ZZ, $Z\gamma$, $\gamma\gamma$, gluon pairs); a bounded vector contact portal couples Ψ to the fermion currents.
8. **The dark sector.** The lightest odd state of Ψ_B, a Dirac fermion of mass $3.1022\,m = 44.3$ TeV ($m = H$), is electrically and weakly neutral at tree level (exactly), stable by the fold parity, cold, and a dark-matter candidate. No state of Ψ is dark energy; the dark energy of UM-2 is a tuned four-dimensional cosmological term.
9. **Cosmology.** The four-dimensional effective theory of the stabilised block is exact general relativity, so the CMB, the CNB and the BAO follow from the standard Einstein-Boltzmann equations with the UM-2 particle content.

3 Space, coordinates and the periodic hidden direction

3.1 The coordinates (AUTHOR)

The author's coordinates are $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$ (the author's notebook: `X = {x0[], x1[], ..., x7[]}`), with

- x_0: the hidden space-like direction, $z \equiv 6Hx_0$ takes every real value;
- x_1, x_2, x_3: ordinary 3-space;
- x_4: the time;
- x_5, x_6, x_7: three extra time-like directions.

$H > 0$ is the author's constant. Frame indices $a = 0, \ldots, 7$ are aligned with $x_0, \ldots, x_7$, and

$$\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1).$$

3.2 The hidden coordinate y, the fold and the tips (DERIVED)

On every interval of z on which $\sin z \neq 0$ define

$$y = \frac{\ln|\sin z|}{6H} \in (-\infty, 0], \qquad |\sin z| = e^{6Hy}, \qquad dy = \cot z \, dx_0.$$

One period $0 < z < \pi$ of x_0 (length $\ell_0 = \pi/(6H)$) consists of two halves, $0 < z < \pi/2$ and $\pi/2 < z < \pi$, joined at the **fold** $z = \pi/2$ ($y = 0$, where $\cos z = 0$). Each half ends at a **tip** ($z \to 0^+$ or $z \to \pi^-$, $y \to -\infty$), at infinite proper distance from the fold. As in the Kohn-Sham record of this repository the tip is replaced by a regular cutoff at $y = -L$ (ASSUMED), with

$$\eta \equiv HL = 3.$$

The fold reflection

$$R: \quad z \mapsto \pi - z \qquad (x_0 \mapsto \ell_0 - x_0)$$

maps the two halves onto each other; modes of a field on one period are even or odd under R. Every function of $y(z)$ is periodic in x_0 with period $\ell_0 = \pi/(6H)$.

4 The metric as a power series

The metric is a series of terms of decreasing magnitude,

$$g_{MN} = \bar{g}_{MN} + \epsilon\, h^{(1)}_{MN} + \epsilon^2 h^{(2)}_{MN} + \ldots, \qquad \epsilon \sim 10^{-5},$$

where ϵ is the size of the CMB temperature anisotropy (STANDARD).

4.1 Order 0: the corrected primordial metric (AUTHOR coordinates, DERIVED properties)

In the author's chart, with two general functions of the time $a_4(x_4)$ and $b_4(x_4) > 0$,

$$\bar{g} = \mathrm{diag}\Big(e^{-2a_4} b_4 \cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6}\Big),$$

where $(\sin^2 z)^{1/6}$ is the positive branch. This is the metric of the Revision record of 2026-10-16 (version 2), whose sources are all diagonal. The notebook metric

$$g^{\mathrm{nb}} = \mathrm{diag}\big(\cot^2 z,\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ -1,\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6}\big),$$

to whose extra-time components the author's order of 2026-10-14 added the factor b_4, differs from it in g_{00} and in the 3-space warp. Define

$$f = e^{-a_4} b_4^{1/2}, \qquad k = \frac{1}{f^2} = \frac{e^{2a_4}}{b_4}, \qquad A = \dot{a}_4, \qquad h = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4,$$

dots being d/dx_4. In the y chart the metric is

$$d\bar{s}^2 = e^{2a_4}(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 + f^2\big[dy^2 - e^{2Hy}(dx_5^2 + dx_6^2 + dx_7^2)\big].$$

The block $\{x_0, x_5, x_6, x_7\}$ is f^2 times a four-dimensional space of constant curvature $-H^2$ and signature $(1,3)$; f is its common scale factor ("the block"), A is the observer's Hubble rate and $h < 0$ means that the extra times and x_0 deflate. The volume element and the vielbein are

$$\sqrt{|\bar{g}|} = e^{3a_4} f^4\, |\cot z|\, (\sin^2 z)^{1/4} = e^{3a_4} f^4 e^{3Hy} \Big|\frac{dy}{dx_0}\Big|,$$

$$e^0{}_{x_0} = f \cot z, \quad e^i{}_{x_i} = e^{a_4}\ (i = 1,2,3), \quad e^4{}_{x_4} = 1, \quad e^t{}_{x_t} = f(\sin^2 z)^{1/12}\ (t = 5,6,7).$$

Curvature (DERIVED; sympy, identical in both charts, 25/25 checks). The Riemann tensor is pair-diagonal, $R^{ab}{}_{cd} = 0$ unless $\{a,b\} = \{c,d\}$, and every sectional curvature $K_{ab} = R^{ab}{}_{ab}$ is one of

$$K_{ij} = A^2, \quad K_{i4} = A^2 + \dot{A}, \quad K_{iB} = Ah, \quad K_{4B} = h^2 + \dot{h}, \quad K_{BB'} = h^2 - H^2 k,$$

with $i,j \in \{1,2,3\}$ and $B, B' \in \{0,5,6,7\}$. None depends on x_0; the curvature is the same on both halves of every period.

4.2 Order 1: the observable perturbations and the hidden tower

Every field and every metric perturbation is expanded in the eigenfunctions $u_n(x_0)$ of its x_0 operator (section 6),

$$h^{(1)}_{MN}(x^\mu, x_0) = \sum_{n\geq 0} h^{(1,n)}_{MN}(x^\mu)\, u_n(x_0).$$

The $n = 0$ term has the flat profile $u_0 = \text{const}$; in the Newtonian gauge its line element is

$$ds^2 = e^{2a_4}\big[(1-2\epsilon\Psi_N)\delta_{ij} + \epsilon h^{TT}_{ij}\big]dx_i dx_j - (1+2\epsilon\Phi)dx_4^2 + f^2(1+2\epsilon\sigma)\big[dy^2 - e^{2Hy}dx_t^2\big],$$

with the two scalar potentials Φ, Ψ_N, the transverse-traceless tensor h^{TT}_{ij} and the radion σ (the relative fluctuation of the block size). The scalar and tensor sectors carry the CMB, the CNB and the BAO; the radion is massive (section 8.3) and does not propagate at cosmological distances. The terms $n \geq 1$ are Kaluza-Klein modes with masses above the gap $3H/(2f)$ (section 6). Order 2 contains the products of order-1 terms and the back-reaction of x_0-localised sources; it is not computed in this version (OPEN).

5 The action

$$\begin{aligned} S = {} & S_{EL}[g] + S_{stab}[g] + S_\Psi[g,\Psi] + S_{gauge}[g,G,W,B] + S_f[g,\psi;G,W,B] \\ & + S_{tip}[g;W,B,\Sigma,\psi] + S_{portal}[\Psi;\psi,G,W,B,\Sigma]. \end{aligned}$$

5.1 Einstein-Lovelock gravity (AUTHOR weights)

$$S_{EL} = \frac{1}{2\kappa}\int d^8x\,\sqrt{|g|}\,\Big(\sum_{k=1}^{3}\alpha_k\,\mathcal{L}_{(k)} - 2\Lambda\Big), \qquad \mathcal{L}_{(k)} = \frac{1}{2^k}\,\delta^{a_1b_1\cdots a_kb_k}_{c_1d_1\cdots c_kd_k}\,R^{c_1d_1}{}_{a_1b_1}\cdots R^{c_kd_k}{}_{a_kb_k},$$

with the generalised Kronecker delta $\delta^{\cdots}_{\cdots}$ (the determinant of ordinary deltas). $\mathcal{L}_{(1)} = R$, $\mathcal{L}_{(2)} = R^2 - 4R_{ab}R^{ab} + R_{abcd}R^{abcd}$ (Gauss-Bonnet) and $\mathcal{L}_{(3)}$ is the cubic Lovelock density. The field equations are

$$\sum_{k=1}^{3}\alpha_k\,E_{(k)}{}^M{}_N + \Lambda\,\delta^M_N = \kappa\,T^M{}_N, \qquad E_{(k)}{}^M{}_N = -\frac{1}{2^{k+1}}\,\delta^{Ma_1b_1\cdots a_kb_k}_{Nc_1d_1\cdots c_kd_k}\,R^{c_1d_1}{}_{a_1b_1}\cdots R^{c_kd_k}{}_{a_kb_k},$$

$E_{(1)}$ being the Einstein tensor. The author's weights are

$$\alpha_k = w_k H^{2-2k}, \qquad w = (w_1,w_2,w_3) = \Big(1, \frac{1}{32}, \frac{1}{1152}\Big), \qquad \Lambda = -10H^2$$

(the author's notebook writes $\Lambda = 10$ with the opposite sign convention for Λ). UM-2 keeps these values.

For a pair-diagonal curvature (DERIVED). When $R^{ab}{}_{cd}$ is pair-diagonal (the order-0 metric),

$$\mathcal{L}_{(k)} = 2^k k! \sum_{\text{k disjoint pairs}}\ \prod_{\text{pairs}} K_{ab}, \qquad E_{(k)}{}^h{}_h = -2^{k-1}k! \sum_{\text{k disjoint pairs avoiding } h}\ \prod_{\text{pairs}} K_{ab},$$

which was checked against the Einstein tensor from the Ricci tensor ($k = 1$), against the explicit Gauss-Bonnet tensor ($k = 2$), against the Bianchi identity $\nabla_M E_{(k)}{}^M{}_N = 0$ ($k = 1,2,3$), against the Euler-Lagrange equations of the action with a lapse ($k = 1,2,3$) and against the record's equations (all exact).

5.2 The block stabiliser (ASSUMED)

A source term

$$S_{stab} = -\int d^8x\,\sqrt{|g|}\,\rho_b(f), \qquad \kappa\,\rho_b(f) = \kappa\rho_b^*\Big(\frac{f_*}{f}\Big)^8,$$

whose energy-momentum tensor is that of a vacuum energy of the block,

$$T^M{}_N[\text{stab}] = \text{diag}\big(p_0,p_3,p_3,p_3,-\rho_b,p_t,p_t,p_t\big), \qquad p_3 = -\rho_b, \qquad p_0 = p_t = +\rho_b.$$

It is conserved ($\dot\rho_b = -3A(\rho_b+p_3) - 4h(\rho_b+p_0) = -8h\rho_b$ gives $\rho_b \propto f^{-8}$), it is four-dimensionally Lorentz invariant ($p_3 = -\rho_b$) and isotropic in the block ($p_0 = p_t$), as the field equations require (section 8.1), and it does not depend on x_0. The f^{-8} law is that of an eight-dimensional energy density whose only scale is the size of the block (a Casimir-type or flux energy). Its normalisation is fixed by the static block (section 8.2): $\kappa\rho_b^* = (26+4\sqrt{51})H^2$ at $f_* = k_*^{-1/2}$. Its microscopic origin is OPEN (a Casimir energy of the fields of the cell or a flux are natural candidates; neither was computed in UM-2).

5.3 The dirac16complex field

5.3.1 The Clifford algebra of the author (AUTHOR, DERIVED verification)

The author's real 16×16 matrices are built as follows ($h = 1,2,3$; $p,q = 1,\dots,4$; ϵ the Levi-Civita symbol; I_n the identity):

$$Q_a(h,p,q) = \epsilon_{hpq4}, \quad Q_b(h,p,q) = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \quad s_h = Q_a - Q_b, \quad t_h = Q_a + Q_b,$$

$$\zeta_0 = I_8, \quad \zeta_{7-h} = \begin{pmatrix} 0 & t_h \\ -t_h & 0 \end{pmatrix}, \quad \zeta_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad \zeta_7 = \zeta_1\zeta_2\zeta_3\zeta_4\zeta_5\zeta_6, \quad \sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix},$$

$$\tau_0 = I_8, \quad \tau_A = \zeta_A \ (A = 1..7), \quad \bar\tau_A = \sigma\,\tau_A^T\sigma, \qquad \gamma^{x_A} \equiv T16[A] = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix} \quad (A = 0,\dots,7).$$

They satisfy (DERIVED, exact; 31/31 checks)

$$\{\gamma^a,\gamma^b\} = 2\eta^{ab} I_{16}, \qquad \tfrac12(\tau_A\bar\tau_B + \tau_B\bar\tau_A) = \eta_{AB} I_8, \qquad \sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7.$$

The charge-conjugation matrix, the chirality, the Krein metric and the generators are

$$C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3} = \begin{pmatrix} -\sigma & 0 \\ 0 & \sigma \end{pmatrix}, \quad \Gamma = \gamma^{x_0}\gamma^{x_1}\cdots\gamma^{x_7} = \mathrm{diag}(-I_8, I_8), \quad B = -iC\gamma^{x_4}, \quad S^{ab} = \tfrac14[\gamma^a,\gamma^b],$$

with $C^T = C$, $C^2 = 1$, $(C\gamma^a)^T = -C\gamma^a$, $B^\dagger = B$, $B^2 = 1$ and B of signature $(8,8)$. The 16-dimensional representation is irreducible under Pin(4,4) (commutant of dimension 1), the double cover of $O(4,4)$, and it splits under Spin(4,4), the double cover of $SO(4,4)$, into two inequivalent irreducible 8-dimensional representations, the eigenspaces $\Gamma = \mp 1$ (commutant of dimension 2, intertwiner of dimension 0).

5.3.2 The Lagrangian and the canonical spin connection

Ψ has 16 complex anticommuting (Grassmann) components and $\bar\Psi = \Psi^\dagger C$. The Lagrangian density is

$$\mathcal{L}_\Psi = \sqrt{|g|}\Big[\tfrac12\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m\,\bar\Psi\Psi - \tfrac{\lambda}{2}(\bar\Psi\Psi)^2\Big],$$

$$\gamma^M = e_a{}^M\gamma^a, \quad D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \quad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \quad \Omega_M = \tfrac12\,\omega_{Mab}S^{ab},$$

$$\omega_M{}^a{}_b = e^a{}_N\big(\partial_M e_b{}^N + \Gamma^N{}_{ML}\,e_b{}^L\big), \qquad \omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b,$$

the canonical (metric-compatible, torsion-free) spin connection: it makes γ^N covariantly constant, $\partial_M\gamma^N + \Gamma^N{}_{ML}\gamma^L + [\Omega_M,\gamma^N] = 0$ (DERIVED). For the order-0 metric (DERIVED, exact, for every a_4, b_4 and on both halves)

$$\gamma^M\Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4},$$

which vanishes only in flat 4+4 space ($H = 0$ with constant a_4, b_4): the coupling of Ψ to gravity is non-trivial. The field equation is

$$\gamma^M D_M\Psi = \big(m + \lambda\,\bar\Psi\Psi\big)\Psi.$$

The mass parameter is the author's $m = 3HM$; UM-2 uses $M = 1/3$, i.e. $m = H$ (ASSUMED, the value of the Kohn-Sham record). λ (dimension mass^{-6}) is the contact self-coupling.

5.3.3 The two halves Ψ_A and Ψ_B and the fold parity (DERIVED)

B commutes with every four-dimensional Lorentz generator $\gamma^a\gamma^b$ ($a,b\in\{x_1,..,x_4\}$) and with C, so

$$\Psi=\Psi_A+\Psi_B,\qquad B\Psi_A=-\Psi_A,\qquad B\Psi_B=+\Psi_B$$

splits Ψ into two four-dimensionally Lorentz-invariant halves of 8 components, and $\bar\Psi\Psi=\bar\Psi_A\Psi_A+\bar\Psi_B\Psi_B$ does not mix them. In the good sector (no extra-time momenta) every interaction of UM-2 commutes with B (DERIVED with the author's matrices: B commutes with C, γ^a and $C\gamma^a$ for $a=x_0,\dots,x_4$ and anticommutes with them for $a=x_5,x_6,x_7$), so the numbers N_A and N_B are separately conserved; extra-time momenta would mix the halves. Under the fold reflection R the modes are even or odd; the bulk Lagrangian is R-symmetric.

5.4 The Standard-Model gauge fields (ASSUMED structure, STANDARD content)

The gauge group is $SU(3)_c\times SU(2)_L\times U(1)_Y$ with eight-dimensional gauge fields G^α_M ($\alpha=1..8$), W^a_M ($a=1,2,3$) and B_M:

$$S_{gauge}=-\frac14\int d^8x\sqrt{|g|}\,g^{MP}g^{NQ}\Big(\frac{1}{\hat g_s^2}G^\alpha_{MN}G^\alpha_{PQ}+\frac{1}{\hat g^2}W^a_{MN}W^a_{PQ}+\frac{1}{\hat g'^2}B_{MN}B_{PQ}\Big),$$

$$G^\alpha_{MN}=\partial_MG^\alpha_N-\partial_NG^\alpha_M+f^{\alpha\beta\gamma}G^\beta_MG^\gamma_N,\quad W^a_{MN}=\partial_MW^a_N-\partial_NW^a_M+\epsilon^{abc}W^b_MW^c_N,\quad B_{MN}=\partial_MB_N-\partial_NB_M.$$

The four-dimensional couplings are $g_s^2=\hat g_s^2/V_\perp$, $g^2=\hat g^2/V_\perp$, $g'^2=\hat g'^2/V_\perp$, where $V_\perp$ is the proper volume of the block per unit 4-volume (section 7). In this geometric normalisation the couplings stand in front of the kinetic terms and not in the covariant derivatives; the dimensions are $[G_M]=[W_M]=[B_M]=1$, $[\hat g^2]=-4$ (mass units), and the four-dimensional couplings are dimensionless. From here on $W^\pm$, Z, A and every formula that shows $\hat g$, $\hat g'$ explicitly refer to the canonically normalised fields $W^a/\hat g$, $B/\hat g'$, $G^\alpha/\hat g_s$ (written with the same letters). The physical fields are

$$W^\pm_M=\frac{W^1_M\mp iW^2_M}{\sqrt2},\qquad Z_M=\cos\theta_W\,W^3_M-\sin\theta_W\,B_M,\qquad A_M=\sin\theta_W\,W^3_M+\cos\theta_W\,B_M,\qquad \tan\theta_W=\frac{g'}{g}.$$

W^+ carries the electric charge $+e$ and W^- the charge $-e$ ($[T_3,T_\pm]=\pm T_\pm$, $Y(W)=0$; W^- is the antiparticle of W^+); Z and the photon A are neutral (DERIVED). A fermion of weak isospin T_3 and hypercharge Y couples to A with $e(T_3+Y)=eQ$, $e=gg'/\sqrt{g^2+g'^2}$, and to Z with $\sqrt{g^2+g'^2}\,(T_3-Q\sin^2\theta_W)$ (DERIVED). The non-abelian terms of $W^a_{MN}W^a_{PQ}$ contain the vertices $W^+W^-\gamma$ (strength e), W^+W^-Z (strength $g\cos\theta_W$) and the quartic $W^+W^-W^+W^-$, $W^+W^-\gamma\gamma$, $W^+W^-Z\gamma$, W^+W^-ZZ; every vertex conserves electric charge, so a single W^+ or W^- always comes with a charged partner.

5.5 The Standard-Model fermions (one generation; ASSUMED structure, INVENTED boundary assignment)

Each multiplet is an eight-dimensional spinor of the same Clifford algebra (16 components times its gauge indices):

field	$SU(3)_c$	$SU(2)_L$	Y	electric charges $Q=T_3+Y$	4D zero mode
$Q=(u,d)_L$	3	2	1/6	u: 2/3, d: $-1/3$	$\Gamma_4=+1$
u_R	3	1	2/3	2/3	$\Gamma_4=-1$
d_R	3	1	$-1/3$	$-1/3$	$\Gamma_4=-1$
$L=(\nu,e)_L$	1	2	$-1/2$	ν: 0, e: -1	$\Gamma_4=+1$
e_R	1	1	-1	-1	$\Gamma_4=-1$
ν_R	1	1	0	0	$\Gamma_4=-1$

with the action

$$S_f=\sum_\psi\int d^8x\sqrt{|g|}\Big[\tfrac12\big(\bar\psi\gamma^MD_M\psi-(D_M\bar\psi)\gamma^M\psi\big)-\mu_\psi\bar\psi\psi\Big],\qquad D_M=\partial_M+\Omega_M-iG^\alpha_MT^\alpha-iW^a_M\frac{\tau^a}{2}-iYB_M,$$

the same canonical spin connection Ω_M as for Ψ and a constant bulk mass μ_ψ per multiplet. The 4D chirality is $\Gamma_4 = i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$.

Boundary conditions. In 16-component form, with the selection projector

$$P_{sel} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-i\gamma^{x_5}\gamma^{x_6}}{2},$$

which commutes with every four-dimensional Lorentz generator (DERIVED), the conditions at the fold ($y=0$) and at the tip ($y=-L$) are

- doublets (Q, L; $\mu_\psi > 0$): fold $P_{sel}(1-\gamma^{x_0})\psi = 0$ and $(1-P_{sel})(1+\gamma^{x_0})\psi = 0$; tip $(1-\gamma^{x_0})\psi = 0$;
- singlets (u_R, d_R, e_R, ν_R; $\mu_\psi < 0$): fold $P_{sel}(1+\gamma^{x_0})\psi = 0$ and $(1-P_{sel})(1-\gamma^{x_0})\psi = 0$; tip $(1+\gamma^{x_0})\psi = 0$.

These are self-adjoint boundary conditions of the Dirac operator (DERIVED). The result (DERIVED, section 6): every doublet field has exactly one four-dimensional Weyl zero mode with $\Gamma_4 = +1$ and every singlet field exactly one with $\Gamma_4 = -1$, all localised at the fold and all in the two blocks selected by P_{sel}, which lie in the $B = -1$ half. The zero-mode spectrum is one Standard-Model generation plus ν_R; its gauge anomalies ($SU(3)^3$, $SU(3)^2U(1)$, $SU(2)^2U(1)$, $U(1)^3$, gravitational $U(1)$, and the $SU(2)$ Witten anomaly) all cancel (DERIVED).

QCD (STANDARD). The gluons are the massless x_0 zero modes of G_M^α. Below the QCD scale the u and d quarks are confined into protons (uud, charge $+1$) and neutrons (udd, charge 0), whose masses $m_p = 938.3$ MeV and $m_n = 939.6$ MeV come from QCD; they are inputs of the cosmology. The QCD scale depends on the block through $g_s^2 = \hat{g}_s^2/V_\perp$.

5.6 The tip: electroweak symmetry breaking without a Higgs particle (INVENTED for this geometry)

At the regular tip a non-linear sigma field $\Sigma(x) \in SU(2)$, $\Sigma^\dagger\Sigma = 1$, transforming as $\Sigma \to U_L\,\Sigma\,U_Y^\dagger$ with $U_Y = \exp(i\theta\tau^3/2)$, carries the electroweak symmetry breaking:

$$S_{tip} = \int_{y=-L} d^7x\,\sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \big(\bar{L}\,\Sigma\,\Lambda_\ell\,\ell_R + \bar{Q}\,\Sigma\,\Lambda_q\,q_R + \mathrm{h.c.}\big)\Big],$$

$$D_\mu\Sigma = \partial_\mu\Sigma - iW_\mu^a\frac{\tau^a}{2}\Sigma + iB_\mu\,\Sigma\frac{\tau^3}{2}, \qquad \ell_R = (\nu_R, e_R), \quad q_R = (u_R, d_R), \quad \Lambda_\ell = \mathrm{diag}(\lambda_\nu, \lambda_e), \quad \Lambda_q = \mathrm{diag}(\lambda_u, \lambda_d).$$

Σ has no radial mode, so there is no Higgs particle; its three would-be Goldstone modes become the longitudinal W^+, W^- and Z. Every term is $SU(2)_L \times U(1)_Y$ invariant (DERIVED with random group elements, deviation 4×10^{-15}). In the unitary gauge $\Sigma = 1$ the first term is

$$-\frac{v_\Sigma^2}{8}\Big[\hat{g}^2\big(W_\mu^1W^{1\mu} + W_\mu^2W^{2\mu}\big) + (\hat{g}^2+\hat{g}'^2)Z_\mu Z^\mu\Big] = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}^2\,W_\mu^+W^{-\mu} + (\hat{g}^2+\hat{g}'^2)Z_\mu Z^\mu\Big],$$

which gives W^+ and W^- one common mass (they are particle and antiparticle), the Z a mass larger by $1/\cos\theta_W$ at leading order (custodial symmetry), and leaves the photon massless. The second term mixes the doublet and singlet zero modes at the tip and gives the fermions their masses (section 6.2).

5.7 The portal interactions of dirac16complex (INVENTED, gauge invariant)

Ψ is a singlet of $SU(3)_c \times SU(2)_L \times U(1)_Y$, so it couples to the Standard Model only through gauge-invariant composite operators. UM-2 contains three of them.

I9: the chiral tip Z portal.

$$S_{I9} = \int_{y=-L} d^7x\,\sqrt{|g_{tip}|}\,\big(c_Z + c'_Z\,\varsigma\big)\,(i\bar{\Psi}\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}\big(\Sigma^\dagger D_\mu\Sigma\,\tau^3\big), \qquad P_+ = \frac{1+\Gamma_4}{2},$$

where $\varsigma = \pm1$ distinguishes the two sides of a tip (the end $z \to n\pi^+$ of one period and the end $z \to n\pi^-$ of the previous one). In the unitary gauge $\frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3) = \frac{v_\Sigma^2}{2}\sqrt{\hat{g}^2+\hat{g}'^2}\,Z_\mu$: the operator couples Ψ to the Z and to nothing else (c_Z, c'_Z have mass dimension -6 on the seven-dimensional tip, where $[v_\Sigma^2] = 5$; only the ratio r below enters the results). Properties (DERIVED):

- P_+ is a Hermitian projector that commutes with the 4D Lorentz generators and with $C\gamma^\mu$, so $i\bar{\Psi}\gamma^\mu P_+\Psi = i(P_+\Psi)^\dagger C\gamma^\mu(P_+\Psi)$ is the (Hermitian) four-vector current of the $\Gamma_4 = +1$ part (with the real matrices $C\gamma^a$ antisymmetric, $\bar{\Psi}\gamma^a\Psi$ is anti-Hermitian and $i\bar{\Psi}\gamma^a\Psi$ Hermitian);
- $\Gamma_4 = -B\gamma^{x_0}$, so $P_+ = (1 - B\gamma^{x_0})/2$: on Ψ_A it is $(1+\gamma^{x_0})/2$, on Ψ_B it is $(1-\gamma^{x_0})/2$, which annihilates Ψ_B at the tip because every orbital satisfies the tip condition $(1-\gamma^{x_0})\Psi = 0$. Hence **no tip operator reaches** Ψ_B: the Ψ_B fold band and the Ψ_B dark matter have zero Z charge and zero transition currents, exactly;
- the Ψ_A fold band (two four-dimensional Weyl species with $\Gamma_4 = +1$) couples to the Z like neutrinos when c_Z is chosen so; the oriented part c'_Z is odd under the fold reflection and lets the heavy odd Ψ_A states decay ($\Psi_A^{odd} \to \Psi_A^{fold} + Z$), while Ψ_B keeps an exact fold parity.

The coupling is fixed by the Z invisible width: the Z-coupling of the Ψ_A fold band is $r = 0.996$ times the neutrino's ($r^2 = 0.992 \pm 0.004$; section 10).

I11: the vector contact portal.

$$S_{I11} = \int d^8x\,\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}\,(i\bar{\Psi}\gamma^M\Psi)\sum_\psi q_\psi\,(i\bar{\psi}\gamma_M\psi),$$

summed over the multiplets with charges q_ψ (each term is separately gauge invariant); $\Lambda_\star$ is the common scale of the two bulk portals I11 and I12, and c_V is dimensionless ($[\bar{\Psi}\gamma^M\Psi] = [\bar{\psi}\gamma_M\psi] = 7$). It couples both halves of Ψ, including the fold bands, to the fermion currents. To keep the sterile Ψ_B fold band out of thermal equilibrium it must be weak: UM-2 sets $c_V = 0$ in the benchmark and computes only the linear response of Ψ to it (section 9).

I12: the gauge-kinetic scalar portal.

$$S_{I12} = -\int d^8x\,\sqrt{|g|}\,\frac{\bar{\Psi}\Psi}{\Lambda_\star^3}\Big[c_G\,G^\alpha_{MN}G^{\alpha MN} + c_W\,W^a_{MN}W^{aMN} + c_B\,B_{MN}B^{MN}\Big]$$

(field strengths in the geometric normalisation of section 5.4; c_G, c_W, c_B dimensionless, since $[\bar{\Psi}\Psi] = 7$ and $[G_{MN}G^{MN}] = 4$). With $W^aW^a = 2W^+_{MN}W^{-MN} + W^3W^3 + \ldots$ and $W^3 = \cos\theta_W Z + \sin\theta_W A$, $B = -\sin\theta_W Z + \cos\theta_W A$, it couples $\bar{\Psi}\Psi$ to the pairs W^+W^-, ZZ, $Z\gamma$, $\gamma\gamma$ and gluon pairs. The scalar density of every chiral zero mode vanishes identically (its lower component is zero), so **I12 does not touch the massless fold bands**; it couples the massive states of Ψ, in particular the dark matter, and gives its relic abundance (section 11). In the QCD vacuum the gluon condensate $\langle G^\alpha G^\alpha\rangle$ makes I12 a uniform mass shift of Ψ.

Not included, and why. A dipole portal $\bar{\Psi}S^{MN}\Psi B_{MN}$ would couple the fold bands to the photon and thermalise the sterile Ψ_B fold band (too much dark radiation). No gauge-invariant operator couples the neutral Ψ to a single W^+ or W^-; $W^\pm$ enter only in pairs (I12) or with a charged fermion pair (the covariant derivatives of I11).

5.8 The catalogue of interactions

#	interaction	fields	equation (vertex)	role in UM-2
1	Einstein-Lovelock self-interaction	g	$\sum_k \alpha_k\mathcal{L}_{(k)} - 2\Lambda$	background, block dynamics, 4D Newton constant $\propto 1/G(k)$
2	gravity of every field	g with all	$\sqrt{\lvert g\rvert}$, g^{MN}, Ω_M in each action	universal coupling; Ψ through the canonical spin connection
3	block stabiliser	g	$-\sqrt{\lvert g\rvert}\rho_b(f)$	static, stable block with attractive 4D gravity
4	Ψ self-interaction	Ψ	$-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$	Kohn-Sham mean fields M_{eff}, v

#	interaction	fields	equation (vertex)	role in UM-2
5	strong interaction	G, u, d	$-G^{\alpha}_{M}\,(i\bar{q}\gamma^{M}T^{\alpha}q)$, $f^{\alpha\beta\gamma}GGG$, $GGGG$	confinement, p and n
6	electromagnetic interaction	A, e, u, d, $W^{\pm}$	$-eQ\,A_M\,(i\bar{\psi}\gamma^{M}\psi)$; $W^{+}W^{-}\gamma$ (strength e)	CMB photons, recombination, Thomson drag
7	weak charged current	$W^{\pm}$, (ν,e), (u,d)	$-\frac{g}{\sqrt{2}}W^{+}_{M}\,i\bar{\nu}\gamma^{M}P_{+}e$, $-\frac{g}{\sqrt{2}}W^{+}_{M}\,i\bar{u}\gamma^{M}P_{+}d$, and h.c.	β decay, $n\leftrightarrow p$, neutrino decoupling
8	weak neutral current	Z, all fermions	$-\sqrt{g^{2}+g'^{2}}\,Z_M$ $\times(T_3-$ $Q\sin^{2}\theta_W)\,i\bar{\psi}\gamma^{M}\psi$	neutrino and Ψ_A decoupling
9	gauge self-interactions	$W^{\pm}$, Z, γ	$W^{+}W^{-}Z$, $W^{+}W^{-}W^{+}W^{-}$, $W^{+}W^{-}\gamma\gamma$, $W^{+}W^{-}Z\gamma$, $W^{+}W^{-}ZZ$	electroweak consistency
10	$W^{\pm}$, Z masses	Σ, W^{a} ($W^{\pm}$, W^{3}), B	$-\frac{v_\Sigma^2}{4}\mathrm{tr}(D\Sigma^{\dagger}D\Sigma)$ at the tip	x_0 profiles bent at the tip
11	fermion masses	Σ, L, ℓ_R, Q, q_R	$\bar{L}\Sigma\Lambda_{\ell}\ell_R+\bar{Q}\Sigma\Lambda_{q}q_R$ at the tip	tip mixing of the chiral zero modes
12	Ψ_A - Z	Ψ_A, Z	I9	Z invisible width, N_{eff}, decay of heavy Ψ_A
13	Ψ - fermion currents	Ψ, all fermions	I11	bounded; Kohn-Sham response
14	Ψ - gauge pairs	Ψ, $W^{+}W^{-}$, ZZ, $Z\gamma$, $\gamma\gamma$, gg	I12	dark-matter relic abundance; QCD-condensate mass shift
15	radion couplings	σ with all massive fields	$\sigma\,m\,\frac{\partial\ln m}{\partial\ln f}\,\bar{\psi}\psi$, etc.	the masses track the block size

6 The x_0 mass mechanism

The author's requirement: mass is given to every non-massless particle by the dependence of its wave function on the space coordinate x_0, periodic with period $\pi/(6H)$; localisation in x_0 implies, through $\Delta x_0\,\Delta p_{x_0}\geq\hbar/2$, a hidden momentum and hence an effective mass. UM-2 implements it exactly: every four-dimensional mass is an eigenvalue of the x_0 operator of its field, acting on wave functions of $y(z)=\ln|\sin z|/(6H)$, $z=6Hx_0$, which are periodic in x_0 with period $\pi/(6H)$. Throughout this section the block is static (f constant), $\xi=y/L\in[-1,0]$ on each half, and eigenvalues are given in units of the proper half cell, $\Lambda=(m\,fL)^2$ for bosons and $e=\varepsilon\,fL$ for fermions.

6.1 Gauge fields and the graviton (DERIVED)

A four-dimensional component $A_\mu(x)\,u(y)$ of a gauge field (gauge $A_y=0$) has the reduced action $\int_{-L}^{0}dy\,f^{4}e^{3Hy}\big[(\partial_y u)^2/f^2-m^2u^2\big]$ per half, hence the x_0 operator and boundary conditions

$$-\frac{1}{f^{2}}\,e^{-3Hy}\,\partial_y\big(e^{3Hy}\partial_y u\big)=m^{2}u\quad\Longleftrightarrow\quad-e^{-3\eta\xi}\big(e^{3\eta\xi}u'\big)'=\Lambda\,u,$$

$$\text{fold: } u'(0)=0\ \text{(even)},\quad u(0)=0\ \text{(odd)};\qquad \text{tip: } u'(-1)=\beta\,u(-1),$$

with $\beta = 0$ for the unbroken generators (photon, gluons) and $\beta > 0$ for the broken ones ($W^{\pm}$, Z; from the Σ term). The transverse-traceless graviton obeys the same operator in Einstein gravity. With $u = e^{-3\eta\xi/2}\tilde{u}$ and $q^2 = \Lambda - 9\eta^2/4$ (primes denote $d/d\xi$) the solutions are elementary and the eigenvalue conditions are

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta\, q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0$$

($q = i\tilde{q}$ below the gap $9\eta^2/4$).

Massless particles. For $\beta = 0$ the flat profile $u = \text{const}$ is an exact eigenfunction with $\Lambda = 0$: the **graviton, the photon and the gluons are exactly massless**; their masslessness is protected by four-dimensional general covariance and gauge invariance. The excited levels are $\Lambda_n = 9\eta^2/4 + n^2\pi^2$ (even, $n \geq 1$) and the roots of $q\cot q = -3\eta/2$ (odd); for $\eta = 3$: $\Lambda_1 = 30.1196$, odd 27.0890, gap 20.25 (DERIVED, finite elements agree to 5×10^{-10}). The Kaluza-Klein scale is $m_{KK} = \sqrt{\Lambda_1}/(fL)$.

$W^{\pm}$ and Z. The Σ term gives β_W for $W^{\pm}$ (one value for both charges) and, by custodial symmetry, $\beta_Z = \beta_W/\cos^2\theta_W$, with θ_W the angle of the gauge couplings, $\cos^2\theta_W = \hat{g}^2/(\hat{g}^2+\hat{g}'^2)$; at small β, $\Lambda \simeq \beta\cdot 3\eta e^{-3\eta}/(1-e^{-3\eta})$ and $m_Z/m_W = 1/\cos\theta_W$. Beyond leading order the bending of the profiles moves the mass ratio $m_W^2/m_Z^2 = \Lambda_W/\Lambda_Z$ (Λ_W, Λ_Z the lowest even eigenvalues for β_W and β_Z) away from $\cos^2\theta_W$: the electroweak parameter

$$\rho = \frac{\Lambda_W}{\Lambda_Z\cos^2\theta_W}$$

exceeds 1. This tree-level custodial violation is what the measured $\rho_0 - 1 = 0.00038$ constrains (ρ_0 measures the violation beyond the Standard-Model loops). With the measured masses, $\rho = m_W^2/(m_Z^2\cos^2\theta_W) = \rho_0$ fixes the angle of the couplings, $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$ (the on-shell value m_W^2/m_Z^2 is 0.776663), and the fit gives (DERIVED)

$$\beta_W = 0.011912, \qquad \Lambda_W = 1.32149 \times 10^{-5}, \qquad m_{KK} = m_W\sqrt{\Lambda_1/\Lambda_W} = 121.3 \text{ TeV},$$

with both masses reproduced: $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W} = 91.1879$ GeV. A positive tip term cannot make $W^{\pm}$ heavier than the Dirichlet limit $\tanh\tilde{q} = 2\tilde{q}/(3\eta)$, $\Lambda_D = 0.010016$, i.e. $m_W \leq 0.01824\, m_{KK}$.

6.2 Fermions (DERIVED)

The eight-dimensional Dirac equation with the canonical spin connection reduces EXACTLY (all eight blocks, general $a_4(x_4)$, $b_4(x_4)$, y-dependent mass $M(y)$ and potential $v(y)$; DERIVED with the author's matrices) under

$$\Psi = e^{ikx_1}\, e^{-\frac{3}{2}Hy}\, e^{a_4/2}\, b_4^{-1}\, V_\beta\, \chi(y, x_4)$$

to $i\partial_4\chi = h_j\chi$ (k the momentum of the mode along x_1) with the 2×2 block Hamiltonian

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big] + v(y),$$

where V_β ($\beta = 0..7$) are the columns of the block basis of the commuting operators $J = \gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$, $K_1 = \gamma^{x_2}\gamma^{x_3}$, $K_2 = \gamma^{x_5}\gamma^{x_6}$ with eigenvalues (j, is_2, is_3). In every block $\gamma^{x_0} = \sigma_3$, $B = js_2$, $C = s_2\sigma_2$ and $\Gamma_4 = -js_2\sigma_3$. The factor $e^{-\frac{3}{2}Hy}$ removes the $\frac{3}{2}(H/f)\gamma^{x_0}$ of the spin connection, and $e^{a_4/2}b_4^{-1}$ removes its γ^{x_4} part. For a constant mass μ and $k = 0$ the general solution of $h_j\chi = 0$ is $\chi = (A\,e^{f\mu y}, B\,e^{-f\mu y})$; the boundary conditions of section 5.5 select one component. With $c = f\mu L$:

- doublet zero mode $\chi = (e^{c\xi}, 0)$, $\Gamma_4 = +1$, localised at the fold for $c > 0$;
- singlet zero mode $\chi = (0, e^{-c\xi})$, $\Gamma_4 = -1$, localised at the fold for $c < 0$;
- at $k \neq 0$: $h_j(e^{c\xi}, 0) = je^{-a_4}k\,(e^{c\xi}, 0)$ exactly: the zero modes are massless four-dimensional particles with $\varepsilon = e^{-a_4}|k|$ (physical momentum).

Masses from the tip. The tip coupling $\bar{L}\Sigma\Lambda_\ell\ell_R$ (and $\bar{Q}\Sigma\Lambda_q q_R$ for the quarks) turns the tip conditions into the self-adjoint mixing condition (DERIVED)

$$b_L(-1) = \kappa_f\, b_R(-1), \qquad a_R(-1) = -\kappa_f\, a_L(-1), \qquad \chi = (a, ib),$$

and the four-dimensional mass is the lowest root of the exact x_0 eigenvalue equation (DERIVED; direct ODE shooting agrees to 3.5×10^{-10})

$$D(e) = e^2 s_L s_R - \kappa_f^2 \,(C_L - c_L s_L)(C_R + c_R s_R) = 0, \qquad s = \frac{\sinh w}{w}, \quad C = \cosh w, \quad w = \sqrt{c^2 - e^2},$$

whose first-order solution is

$$e = \kappa_f \,\zeta(c_L)\,\zeta(-c_R), \qquad \zeta(c)^2 = \frac{c\,e^{-c}}{\sinh c} \simeq 2c\,e^{-2c} \quad (c > 0),$$

i.e. the tip mixing times the probability densities of the two chiral wave functions at the tip ($c_L > 0$ and $c_R < 0$: both zero modes are localised at the fold). With $\kappa_f = 1$ and $c_L = -c_R = c$ the measured masses require (DERIVED; the first-order formula agrees with the exact root to 1.2×10^{-8})

fermion	mass	$c = f\mu L$	μ/H	proper width $fL/(2c)$
ν	0.05 eV	18.671	28.87	2.4×10^{-22} m
e	0.511 MeV	10.304	15.93	4.3×10^{-22} m
u	2.16 MeV	9.545	14.76	4.7×10^{-22} m
d	4.70 MeV	9.135	14.13	4.9×10^{-22} m

Order-one bulk masses give the whole hierarchy, because the tip densities depend exponentially on c. The neutrino is a Dirac fermion; its right-handed partner ν_R is a gauge singlet that is never thermalised.

6.3 The quanta of dirac16complex (DERIVED)

Ψ has the same block Hamiltonian with $M = m$ and the boundary conditions fold $(1 - \gamma^{x_0})\Psi = 0$ (even) or $(1 + \gamma^{x_0})\Psi = 0$ (odd) and tip $(1 - \gamma^{x_0})\Psi = 0$; both parities are modes of the full period. With $c_\psi = fmL$ the exact $k = 0$ spectrum is

- even: the zero mode $\chi = (e^{c_\psi \xi}, 0)$, $\varepsilon = 0$, in all eight blocks, and $\varepsilon = \pm\sqrt{m^2 + (n\pi/(fL))^2}$;
- odd: $\varepsilon = \pm\sqrt{m^2 + p^2}$ with $\tan(p\,fL) = -p/m$.

The **fold band** $\varepsilon = e^{-a_4}|k|$ is massless: 8 states per 3-momentum, 4 in Ψ_A (two Weyl species with $\Gamma_4 = +1$) and 4 in Ψ_B (two Weyl species with $\Gamma_4 = -1$). The **bulk band** is massive; at the stabilised block ($f_* = 0.21556$, $c_\psi = 0.64667$) its lowest level is the odd level $\varepsilon_{DM} = 3.10220\,m$ ($p\,fL = 1.89901$, $p = 2.93661\,m$), and the lowest even bulk level is $4.95998\,m$. The odd level is the dark-matter candidate (section 11).

6.4 What the uncertainty relation does and does not bound (DERIVED)

In the flat measure $u = e^{-3\eta\xi/2}\tilde{u}$ the exact identity $e^{3\eta\xi}u'^2 = \tilde{u}'^2 + \frac{9}{4}\eta^2\tilde{u}^2 - \frac{3}{2}\eta\,(\tilde{u}^2)'$ turns every bosonic eigenvalue into

$$(m\,fL)^2 = \langle p^2\rangle + \frac{9\eta^2}{4} - \frac{3\eta}{2}\,\varrho(\text{fold}) + \Big(\frac{3\eta}{2} + \beta\Big)\varrho(\text{tip}), \qquad \langle p^2\rangle = \frac{\int \tilde{u}'^2}{\int \tilde{u}^2}, \quad \varrho = \frac{\tilde{u}^2}{\int \tilde{u}^2},$$

and every fermionic one into $(\varepsilon fL)^2 = \langle|\chi'|^2\rangle + c^2 - c\,[\chi^\dagger\sigma_3\chi]_{-1}^0/\int|\chi|^2$, with $\langle p^2\rangle = \langle|\chi'|^2\rangle$ and $\varrho = |\chi|^2/\int|\chi|^2$ (integrals over $\xi \in [-1, 0]$, primes $d/d\xi$). The proper uncertainty relation bounds only the hidden kinetic energy. On the cell, which ends at the tips, it reads

$$\sqrt{\langle p^2\rangle}\,\Delta\xi \geq \frac{1 - \varrho(\text{tip})}{2}, \qquad \Delta\xi^2 = \int_{-1}^{0} \xi^2\,\varrho\,d\xi,$$

since $1 = \int \varrho\,d\xi = \varrho(\text{tip}) - \int \xi\varrho'\,d\xi$ (the fold end $\xi = 0$ contributes nothing) and the Cauchy-Schwarz inequality bounds $|\int \xi\varrho'\,d\xi|$ by $2\Delta\xi\sqrt{\langle p^2\rangle}$; $\Delta s = fL\,\Delta\xi$ is the root-mean-square proper distance from the fold. The familiar $\langle p^2\rangle \geq 1/(4\Delta\xi^2)$ is guaranteed only for profiles negligible at the tip (it can hold otherwise: the first even Kaluza-Klein mode, with $\varrho(\text{tip}) = 0.655$, has $\langle p^2\rangle\Delta\xi^2 = 4.59$); the photon zero mode has 0.497, the broad fold zero mode of Ψ ($\varrho(\text{tip}) = 0.489$) only 0.097. The relation with the tip term is a theorem; it is also checked numerically for each of the eleven computed states (the zero mode of the photon, gluons and graviton, $W^\pm$, Z, the first even and odd Kaluza-Klein modes, the ν, e, u, d zero modes, the fold zero mode of Ψ and the dark matter; DERIVED). Three classes follow:

- **massless**: profiles flat relative to the geometry ($u = \text{const}$ or $h\chi = 0$): the fold term cancels the kinetic energy and the warp exactly; graviton, photon, gluons and the Ψ fold band. They are localised at the fold ($\Delta s = 1.4 \times 10^{-21}$ m for the photon) and yet massless;
- **heavy**: the x_0 dependence fills the cell. For the gauge fields and the graviton every odd mode and every excitation has $m \geq 3H/(2f) = 99.5$ TeV (the gap theorem of the boson operator: one even level below the gap for every $\beta \geq 0$, no odd level); the Ψ excitations start at its lowest odd level, $3.10m = 44.3$ TeV (the dark matter), since the Dirac operator has no warp gap. For the computed states of this class the author's estimate $\hbar c/(2\Delta s)$, with Δs the root-mean-square proper distance from the fold, lies below the mass (by factors 2.3 to 7.5, the dark matter included); this is observed for these states, not derived for all;
- **light**: the x_0 dependence that costs mass sits at the tip, where the probability is tiny: $W^{\pm}$, Z, e, ν, u, d. Their masses lie 10^{-3} ($W^{\pm}$) to 10^{-16} (ν) below $\hbar c/(2\Delta s)$, and a sharper localisation gives a LIGHTER fermion ($d\ln m_f/d\ln c = -17$ to -36).

7 Dimensional reduction and the four-dimensional constants

The good sector (no dependence on x_5, x_6, x_7) is kept (ASSUMED): modes with extra-time momentum q have $\omega^2 = M^2 + k^2 - q^2/f^2$ and are tachyonic for $q > fM$ (OPEN). The extra times are compact with coordinate volume v_t (ASSUMED). Integrating over one period of x_0 (both halves) and the extra times, the proper block volume per unit 4-volume is

$$V_\perp = 2\int_{-L}^{0} dy\, f^4 e^{3Hy}\, v_t = \frac{2f^4 v_t\,(1-e^{-3\eta})}{3H},$$

and the reduced action of section 8.5 gives

$$\bar{M}_{Pl}^2 = \frac{G(k)\,V_\perp}{\kappa} = \frac{1}{8\pi G_N}, \qquad g^2 = \frac{\hat{g}^2}{V_\perp}, \quad g'^2 = \frac{\hat{g}'^2}{V_\perp}, \quad g_s^2 = \frac{\hat{g}_s^2}{V_\perp}.$$

($\bar{M}_{Pl}$ is the reduced Planck mass). Every four-dimensional constant depends on the block: α and the gauge couplings $\propto f^{-4}$, $G_N \propto f^{+2.05}$ ($d\ln G_N/d\ln f = -4 + 2kG'(k)/G(k)$ at k_*: the Lovelock factor varies too), Kaluza-Klein masses $\propto f^{-1}$, $m_W \propto f^{-0.50}$, $m_e \propto f^{-20.6}$, $M_{DM} \propto f^{-0.77}$ (DERIVED). The masses are constant only because the block is stabilised.

8 Gravity: the background, the stabilised block, the deflating extra times

8.1 The background field equations (DERIVED)

For the order-0 metric every off-diagonal component of every $E_{(k)}$ vanishes and the hidden and extra-time components coincide, so the source must be diagonal, independent of x_0 and block-isotropic:

$$T^M{}_N = \mathrm{diag}(p_0, p_3, p_3, p_3, -\rho, p_t, p_t, p_t), \qquad p_0 = p_t.$$

The three independent components are the x_4 constraint (first order in time derivatives), the 3-space component and the block component; the last two are linear in $(\dot{A}, \dot{h})$. Their complete form follows from the pairing formula of section 5.1 with the sectional curvatures of section 4.1. A source with $p_0 \neq p_t$ (the Kohn-Sham states of Ψ and the matter at the fold have it; the dark matter has $p_0/\rho = 0.772$, $p_t = 0$) acts on the common block scale f through the block-averaged pressure $\bar{p} = (p_0 + 3p_t)/4$ (the variation of the action with respect to f), while $p_0 - p_t$ drives a block-shape mode that the order-0 metric does not contain (OPEN); for matter its size relative to the stabiliser is of order $(A/H)^2$, 10^{-92} today. To first order about the static block of section 8.2 with block-averaged matter sources (A^2 counts as first order: the x_4 constraint gives $3G(k_*)A^2 = \kappa\delta\rho$, because $\frac{d}{dk}[6H^2k(1-2w_2k) + \kappa\rho_b]$ vanishes at k_*), the equations give (DERIVED)

$$\delta\dot{A} = J_{Ak}\,\delta k + s_{A3}\,\kappa\delta p_3 + s_{Ap}\,\kappa\delta\bar{p} + q_A\,\kappa\delta\rho,$$

$$\delta\dot{h} = J_{hk}\,\delta k + s_{h3}\,\kappa\delta p_3 + s_{hp}\,\kappa\delta\bar{p} + q_h\,\kappa\delta\rho, \qquad \delta\dot{k} = -2k_*\,\delta h,$$

$$\delta\ddot{k} + \omega_r^2\,\delta k = -2k_*\big(s_{h3}\,\kappa\delta p_3 + s_{hp}\,\kappa\delta\bar{p} + q_h\,\kappa\delta\rho\big), \qquad \omega_r^2 = 2k_*J_{hk},$$

with $J_{Ak} = \frac{6(238-19\sqrt{51})}{2249} H^2 = 0.272956H^2$, $J_{hk} = \frac{4063-485\sqrt{51}}{2249} H^2 = 0.266522H^2$, $s_{A3} = \frac{2283}{4498} - \frac{162\sqrt{51}}{2249} = -0.006853$, $s_{Ap} = \frac{28\sqrt{51}-114}{2249} = 0.038221$, $q_A = \frac{135021}{265382} - \frac{10308\sqrt{51}}{132691} = -0.045997$, $s_{h3} = \frac{21\sqrt{51}}{2249} - \frac{171}{4498} = 0.028666$, $s_{hp} = \frac{239\sqrt{51}}{6747} - \frac{485}{2249} = 0.037320$, $q_h = \frac{57}{4498} - \frac{7\sqrt{51}}{2249} = -0.009555$. In the adiabatic response ($\delta\ddot{k} = 0$) the block shifts by $\delta k = (-0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p} + 0.035852\,\kappa\delta\rho)/H^2$ and 3-space obeys exactly $\delta\dot{A} = -\kappa(\delta\rho + \delta p_3)/(2G(k_*))$: the four-dimensional Friedmann acceleration equation with $8\pi G_N = \kappa/(G(k_*)V_\perp)$.

8.2 The static block and attractive gravity with the author's weights (DERIVED)

For a static block ($h = 0$, k constant) and $\dot{A} = 0$ the components are

$$x_4: \quad -3G(k)A^2 + 6H^2k(1-2w_2k) + \Lambda + \kappa\rho = 0, \qquad x_1: \quad \text{the same with } -\kappa p_3 \text{ for } \kappa\rho,$$

$$x_0: \quad 6(12w_2k-1)A^2 - \frac{12(w_2-18w_3k)}{H^2}A^4 + 3H^2k + \Lambda - \kappa p_0 = 0, \qquad G(k) = 1 - 24w_2k + 72w_3k^2.$$

$G(k)$ is the coefficient of the four-dimensional Einstein-Hilbert term (section 8.5): gravity is attractive for $G > 0$. Without sources the only static block with $A = 0$ (and $k \neq 0$) is $k_0 = 1/(4w_2) = 8$ with $\Lambda_0 = -24H^2$ and $G(k_0) = -5 + 9w_3/(2w_2^2) = -1$: with the author's weights the sourceless block has REPULSIVE gravity, and the author's $\Lambda = -10H^2$ is not Λ_0. For the author's weights $G(k) = (k^2 - 12k + 16)/16$, positive only for $k < 6 - 2\sqrt{5}$ or $k > 6 + 2\sqrt{5} = 10.47$.

With the block stabiliser ($p_3 = -\rho_b$, $p_0 = p_t = \rho_b$) and $A = 0$ the two conditions require

$$\kappa\rho_b = 10H^2 - 6H^2k + \tfrac{3}{8}H^2k^2 = \kappa p_0 = 3H^2k - 10H^2 \quad \Longrightarrow \quad k_\pm = 12 \pm \tfrac{4}{3}\sqrt{51}.$$

The upper root is the UM-2 vacuum:

$$k_* = 12 + \tfrac{4}{3}\sqrt{51} = 21.5219, \qquad f_* = k_*^{-1/2} = 0.215556,$$

$$G(k_*) = \tfrac{20}{3} + \sqrt{51} = 13.8081 > 0, \qquad \kappa\rho_b^* = (26 + 4\sqrt{51})H^2 = 54.566H^2,$$

with attractive four-dimensional gravity and the author's weights and Λ unchanged. The lower root $k_- = 2.4781$ has $G = -0.4748$.

8.3 Stability: the radion (DERIVED)

Linearising the 3-space and block components (solved for $\dot{A}$, $\dot{h}$) with $\dot{k} = -2kh$ and $\rho_b \propto k^4$ about k_* gives the eigenvalues $0, \pm i\,\omega_r$ with

$$\frac{\omega_r^2}{H^2} = \frac{31552}{2249} - \frac{2416\sqrt{51}}{6747} = 11.4721, \qquad \omega_r = 3.38705\,H:$$

the block is stable and oscillates; the zero eigenvalue is the direction of a small four-dimensional cosmological term. The radion (the quantum of block-size oscillations) has the mass $m_r = 3.387H = 48.4$ TeV. Stability holds for each of the stabiliser laws $p_0 = \nu\rho_b$, $\rho_b \propto k^{2(1+\nu)}$, computed at $\nu = 1/4, 1/2, 3/4, 1, 5/4, 3/2, 2, 3$ (radion masses $3.33H$ to $4.10H$).

8.4 The deflating extra times (DERIVED, full nonlinear equations)

The full nonlinear equations (all three components, the x_4 constraint preserved to 6×10^{-10}) were integrated from an expanding 3-space (A from the constraint) and a block LARGER than the stabilised one ($k_0 < k_*$, i.e. $f_0 > f_*$) that deflates at $h_0 = -H/2, -H/3, -H/10$, for k_0 from 1.5 to 20. Every start with k_0 above the separatrix $k_c(h_0) = 2.6983, 2.5991, 2.5301$ (for $h_0 = -H/2, -H/3, -H/10$; just above the unstable static block $k_- = 2.4781$, whose stable manifold the separatrix is) ends at k_*: x_5, x_6, x_7 and x_0 deflate (at rates up to $|h| = 2.63H$ in the stored runs) by up to the factor $f(k_c)/f_* = (k_*/k_c)^{1/2} = 2.917$, reach k_* for the first time within $x_4 < 1/H$, and then oscillate about k_* with the radion frequency ($3.3868H$ measured, $3.3870H$ linear), while 3-space keeps expanding; the oscillation energy redshifts like dust ($A\,x_4 = 0.667$ to 0.670 late, $2/3$ for dust). Starts with $k_0 < k_c$ (still larger blocks) become singular within $x_4 = 0.2/H$ to $1.4/H$ or admit no expanding initial data. The extra times therefore deflate exponentially only for a short primordial era and are then frozen, which is what the constancy of the masses requires (section 7).

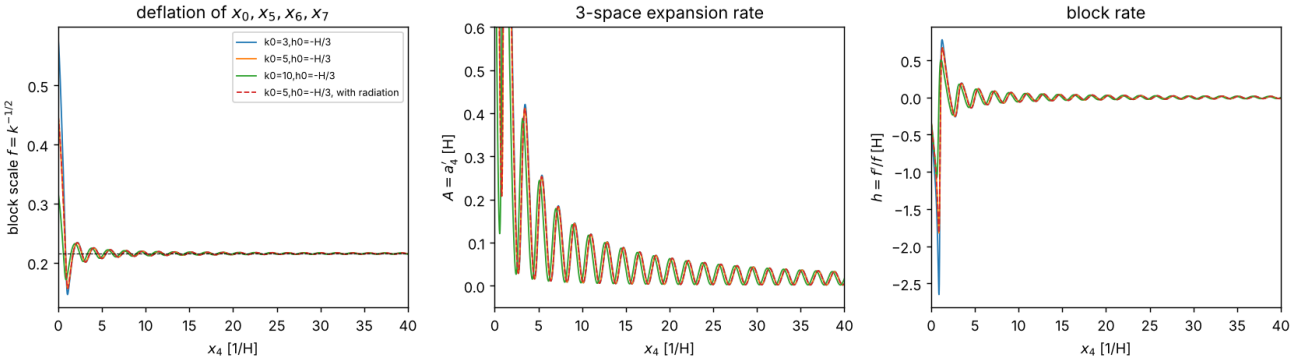


Figure 1: Deflation of the extra times into the stabilised block (full nonlinear Einstein-Lovelock equations with the author's weights).

8.5 The four-dimensional effective gravity (DERIVED)

On a product of a four-dimensional metric with the static block (constant curvature $K = -H^2 k$), the Lovelock densities factorise, $\mathcal{L}_{(p)} = \sum_{r+s=p} \frac{p!}{r!\,s!} \mathcal{L}^{(4)}_{(r)} \mathcal{L}^{(int)}_{(s)}$ ($p = 1, 2, 3$ the Lovelock order) with $\mathcal{L}^{(int)}_{(s)} = (1, 12K, 24K^2, 0)$ for $s = 0, 1, 2, 3$, and $\mathcal{L}^{(4)}_{(3)} \equiv 0$ in four dimensions while $\mathcal{L}^{(4)}_{(2)}$ (the four-dimensional Gauss-Bonnet density) is topological. The reduced Lagrangian (gravity plus the stabiliser, in units of $1/(2\kappa)$) is

$$\sum_p \alpha_p \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)\, R^{(4)} + (\text{topological}) - 2\, G(k)\, \Lambda_4, \qquad \Lambda_4 = \frac{6H^2 k(1 - 2w_2 k) + \Lambda + \kappa\rho_b}{G(k)},$$

so the four-dimensional gravity of UM-2 is EXACTLY general relativity with Newton's constant $8\pi G_N = \kappa/(G(k_*)V_\perp)$ and the cosmological term Λ_4, which vanishes at the static point by construction. The observed dark energy is a small residual of this balance, $\Lambda_4/H^2 \sim 10^{-92}$: tuned (the cosmological-constant problem, OPEN).

9 dirac16complex in the cell: Kohn-Sham description

Ψ is quantised canonically with x_4 as the time, $\{\Psi_r(x), \Psi^\dagger_s(x')\} = B_{rs}\, \delta^7(x - x')/\sqrt{|g|}$ on a slice (r, s spinor indices); the indefinite Krein metric B is handled in the good sector by counting the positive branch of every block as particles (the record's convention, ASSUMED). The interacting many-body state is described by Kohn-Sham theory: orbitals of h_j with the mean fields

$$M_{eff}(y) = m + \tfrac{15}{16}\lambda S(y) + \delta M_{portal}(y), \qquad v(y) = -\tfrac{1}{16}\lambda\, n(y) + v_{portal}(y),$$

$$e_{int} = \tfrac{15}{32}\lambda S^2 - \tfrac{1}{32}\lambda n^2, \qquad E_{KS} = \sum_o g f_o \varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} e^{3Hy} e_{int}\, dy,$$

where n and $S = \bar\Psi\Psi$ are the proper densities, $g = 4r_3(|\vec n|^2)$ the degeneracy of the momentum shell $\vec k = \Delta k\, \vec n$ ($\vec n \in \mathbb{Z}^3$, $r_3(|\vec n|^2)$ the number of representations of $|\vec n|^2$ as a sum of three squares, $\Delta k = 0.25H$), $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f^4$ (at $a_4 = 0$) the normalisation volume of a cell, of dimension length^6 (the proper seven-volume of a cell, with its hidden extent, is $2\,\mathrm{Vol}_7(1 - e^{-3\eta})/(3H)$), f_o the occupations (aufbau or Mermin), and the portal mean fields come from I11 (the fermion densities), I12 (the gluon condensate) and I9 (the Z charge at the tip, on Ψ_A only). The energy-momentum tensor of a Kohn-Sham state,

$$\rho = \sum_o w g f_o \varepsilon_o\, n_o - e_{int}, \qquad p_3 = \sum_o w g f_o\, \tfrac{e^{-a_4|k|}}{3} t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum_o w g f_o \big[(\varepsilon_o - v) n_o - M_{eff} s_o - e^{-a_4|k|} t_o\big] + e_{int},$$

with $n_o = P\chi^\dagger_o \chi_o$, $s_o = P\chi^\dagger_o j\sigma_2 \chi_o$, $t_o = P\chi^\dagger_o j\sigma_3 \chi_o$, $P = e^{-3Hy}/\mathrm{Vol}_7$ and $w = 1/2$ (the densities are $n = \sum_o w g f_o n_o$ and $S = \sum_o w g f_o s_o$), is the source of Ψ in the gravitational equations; its block-averaged pressure $(p_0 + 3p_t)/4$ pushes on the block (section 8.1).

10 Neutrinos, Ψ_A, the Z invisible width, N_{eff} and the CNB (DERIVED)

With one generation the Z decays invisibly into $\nu\bar{\nu}$ and into the Ψ_A fold band (two Weyl species),

$$N_\nu \equiv \frac{\Gamma_{inv}(Z)}{\Gamma(Z \to \nu\bar{\nu})} = 1 + 2r^2, \qquad r = \frac{g_Z(\Psi_A)}{g_Z(\nu)};$$

the measured $N_\nu = 2.9840 \pm 0.0082$ gives $r^2 = 0.9920 \pm 0.0041$. The Ψ_A species then interact with the plasma like ν_μ and ν_τ of the three-generation Standard Model and decouple with the neutrinos ($T \approx 2.6$ MeV, after the QCD and pion entropy releases; any $r^2 > 2 \times 10^{-3}$ gives the same temperature), so

$$N_{eff} = 3.044, \qquad T_\nu = T_{\Psi_A} = \left(\frac{4}{11}\right)^{1/3} T_{CMB} = 1.945 \text{ K}.$$

The cosmic neutrino background of UM-2 consists of ν_e (massive, 0.05 eV, non-relativistic today) and two massless Ψ_A species, 112 cm^{-3} per species, 336 cm^{-3} in all. The Ψ_B fold band is never thermalised (no tip coupling, no scalar density, $c_V = 0$) and contributes nothing.

11 The dark sector

Dark matter (CONDITIONAL candidate). The lightest odd state of Ψ_B, a Dirac fermion of mass

$$M_{DM} = \varepsilon_{DM} = 3.10220\, m = 44.3 \text{ TeV} \qquad (m = H = 14.30 \text{ TeV}),$$

is cold ($w = 0$ at rest), electrically neutral and Z-neutral exactly at tree level (no tip operator reaches Ψ_B), stable as long as the fold parity of Ψ_B is exact (it is a symmetry of every interaction that reaches Ψ_B) and N_B is conserved (both hold in the good sector). Its hidden wave number carries 90 % of ε^2: its mass is the author's x_0 mechanism in the Heisenberg sense. A thermal relic through I12 (p-wave annihilation into W^+W^-, ZZ, $Z\gamma$, $\gamma\gamma$, gluon pairs; standard radiation-dominated freeze-out) needs $\langle\sigma v\rangle(x_f) = 1.37 \times 10^{-25}$ cm^3/s, below the s-wave thermal unitarity estimate $(4\pi/M_{DM}^2)\sqrt{x_f/\pi} = 2.3 \times 10^{-25}$ cm^3/s, i.e. $(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (16.6 \text{ TeV})^{-3}$ for the four-dimensional coefficients of $G_i\, \bar{\chi}\chi F^i_{\mu\nu}F^{i\mu\nu}$ (canonical fields), $G_i = \mathcal{O}\, c_i\, g_i^2/\Lambda_\star^3$ with g_i the four-dimensional gauge couplings and $\mathcal{O} = \partial\varepsilon_{DM}/\partial m = 0.7072$ the response of the dark-matter level to a uniform mass shift (DERIVED, Hellmann-Feynman); direct detection is far below current limits. The relic value assumes that the radion oscillation left over from the deflation does not dominate and dilute the Universe after freeze-out, i.e. it is conditional on the solution of the moduli problem (section 14).

Dark energy (NOT SUPPORTED). No positive-energy state of Ψ has an equation of state $w = p_3/\rho < 0$ on the static block; a $\lambda < 0$ condensate tuned to $w_0 = -1$ runs away ($w_a = -dw/da = -6$ at $a = 1$, the coefficient of the form $w = w_0 + w_a(1 - a)$, a the scale factor) and its energy density turns negative at redshift $z = 0.26$. The dark energy of UM-2 is the tuned residual Λ_4.

12 Cosmology: CMB, CNB, BAO

The four-dimensional effective theory is general relativity (section 8.5) with photons, ν_e, the two Ψ_A species, baryons, the Ψ_B dark matter and Λ_4. Its linear perturbations (order ϵ) obey the standard Einstein-Boltzmann equations. Solved with the Einstein-Boltzmann code CAMB at the Planck 2018 values of ω_b, ω_c, τ_{reio} (the optical depth to reionisation), A_s, n_s and θ_* (DERIVED): $H_0 = 67.43$ km/s/Mpc, $r_{drag} = 147.10$ Mpc, the TT and EE spectra agree with Planck ΛCDM to 2.2×10^{-4} and 3.7×10^{-4} ($30 \le \ell \le 2500$; for TT at most 0.011 of the cosmic variance $\sqrt{2/(2\ell+1)}$ of a single multipole), and the DESI DR2 BAO give $\chi^2 = 30.3$ for 13 measurements (Planck ΛCDM: 32.6).

13 Parameters

parameter	value	status
H	14.30 TeV	DERIVED from m_{KK}, η and f_*
Lovelock weights w, Λ	$(1, 1/32, 1/1152)$, $-10H^2$	AUTHOR
$\eta = HL$	3	ASSUMED (record)
$m = 3HM$	$M = 1/3$ $(m = H)$	ASSUMED (record)
stabiliser law	$\rho_b \propto f^{-8}$, $p_0 = p_t = \rho_b = -p_3$	ASSUMED
k_*, f_*, $G(k_*)$	21.522, 0.21556, 13.808	DERIVED
m_r	48.4 TeV	DERIVED
m_{KK}	121.3 TeV	DERIVED (fit to $\rho_0 - 1$ and m_W)
β_W, β_Z; $\cos^2\theta_W$	0.011912, 0.015343; 0.776368	DERIVED (fit; $\beta_Z = \beta_W/\cos^2\theta_W$, θ_W the angle of the couplings, fixed by m_W/m_Z and ρ_0)
κ_f; c_ν, c_e, c_u, c_d	1; 18.671, 10.304, 9.545, 9.135	ASSUMED; DERIVED (fit to the masses)
m_ν	0.05 eV	ASSUMED
r^2 (I9)	0.992	DERIVED (fit to N_ν)
c_V (I11)	0 (benchmark)	ASSUMED
I12 coefficients	$(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (16.6\ \text{TeV})^{-3}$	DERIVED (relic requirement)
M_{DM}	44.3 TeV	DERIVED

14 Assumptions, open problems and conflicts with data

- **ASSUMED**: the regular tip cutoff $\eta = 3$; the good sector and compact extra times; $m = H$; the block stabiliser and its f^{-8} law; the boundary assignments of the Standard-Model fields; the tip sigma field; $\kappa_f = 1$, $c_L = -c_R$; $m_\nu = 0.05$ eV; the record's particle convention for the Krein metric.
- **INVENTED**: the portals I9, I11, I12; the oriented part of I9.
- **OPEN**: the microscopic origin of the stabiliser; the tachyonic extra-time modes outside the good sector; the closed time-like curves implied by compact extra times; the radion oscillation left over from the deflation (ESTIMATE: its Planck-suppressed decay rate $\Gamma = C_r\, m_r^3/(4\pi \bar{M}_{Pl}^2)$, with $\bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV and an order-one coupling factor C_r, gives a lifetime of $0.43\,\text{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma \bar{M}_{Pl}} = 1.85\sqrt{C_r}$ MeV ($g_* = 10.75$), below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: the cosmological moduli problem, which also conditions the dark-matter relic); the Kaluza-Klein graviton spectrum with the Lovelock corrections; the S parameter of the Higgsless tip; anomalies of the eight-dimensional bulk theory; the cosmological constant (Λ_4 tuned to $10^{-92}H^2$); the electroweak sector beyond tree level (the angle of the couplings of the $W^\pm$, Z mass fit, $\sin^2\theta_W = 0.22363$, differs from the $\overline{MS}$ value 0.23122 used in the Z couplings by the Standard-Model loop corrections, which UM-2 does not compute; the tree-level Z widths differ by at most 1.4 % between the two).
- **CONFLICTS WITH DATA, inherent in the order**: there is no Higgs boson while one is observed at 125 GeV; without it the longitudinal W^+W^- scattering violates perturbative unitarity near $\sqrt{8\pi}\,v \approx 1.2$ TeV, far below the Kaluza-Klein modes that could restore it; with one generation the total Z width is 1.23 GeV against the measured 2.4955 GeV (the μ, τ, s, c, b channels are missing), while its invisible part is reproduced through the fit of N_ν.